

Salimli Ayzek

Data Scientist. 1st-year master's student, SPbSTU, ICCS. Mathematics and Computer Science, Machine Learning and Artificial Intelligence.

thebloomsbridge - Work

GitHub

Hugging Face

ORCID

ResearchGate

eLibrary

Saint-Petersburg

Версия на русском 🇷🇺

Download resume (ENG)

Download this portfolio

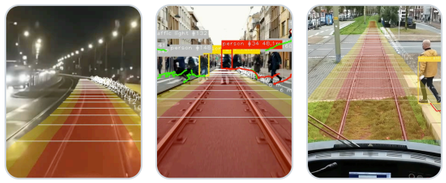
About

- Salimli Ayzek - master's student at SPbSTU (ICCS), majoring in Mathematics and Computer Science: Machine Learning and Artificial Intelligence. I have 1 year and 5 months of experience at The Blooms Bridge: AI systems integration (RAG, computer vision, etc.) and backend development. I primarily position myself as a Data Scientist and work with the ML cycle: data preprocessing and transformation, data analysis, feature selection, model training, hyperparameter tuning, result interpretation, data labeling, LLM training, model building (PyTorch, TensorFlow), deployment (LangChain + LangGraph), and computer vision tasks (YOLO, ResNet, MobileNet, ViT, U-Net, SegFormer, YOLO-seg, YOLO-pose, MediaPipe, OpenCV).
- My professional interests include computer vision, statistical and machine learning algorithms, neural networks, functional programming, optimization methods, statistics, and discrete mathematics.
- Outside work: travel, philosophy (especially how cognitive thinking and foundational ideas connect with computer science), football, and neue deutsche harte.

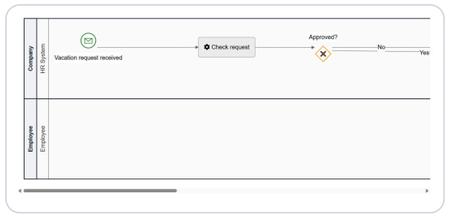
Projects

ML and AI

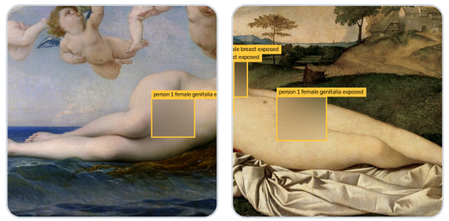
Object detection on dynamic RoI based on segmentation with tracking (weights available)



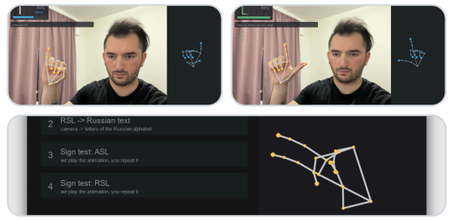
Text-to-BPMN (generating BPMN diagrams from text)



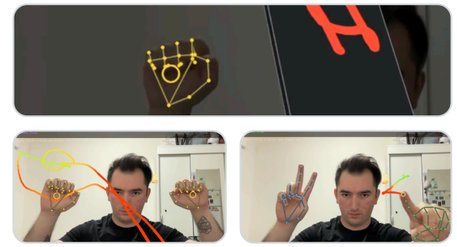
Real-time censorship API (for online meetings)



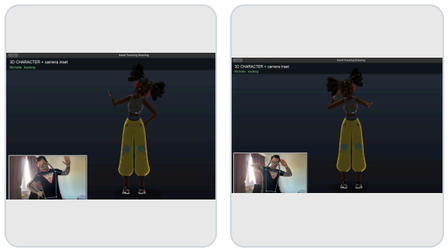
ASL, RSL sign languages to text (text to sign) translator



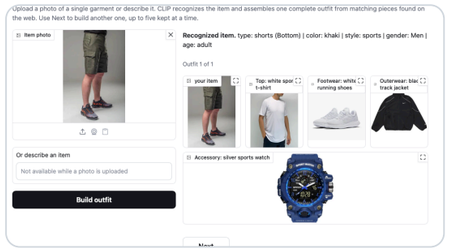
Hand tracking drawing (with 3D)



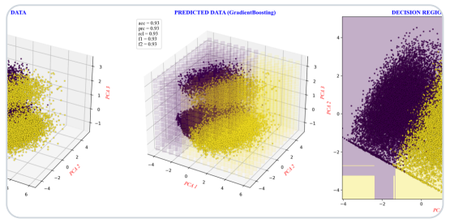
Pose driven 3D characters



Vision fashion recommender



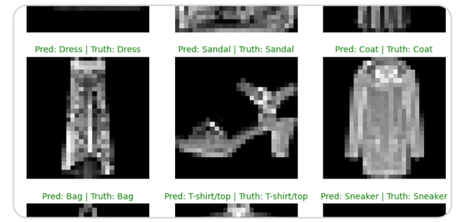
Heart disease classifier



Qt MNIST recognizer



Fashion-MNIST classification with TinyVGG



Sarcasmic text classification (transformer NN)

```
Confidence: 0.9799

Text: The guy found way to end with Global Hungry, but cant find his keys
Prediction: Sarcasm
Confidence: 0.9535

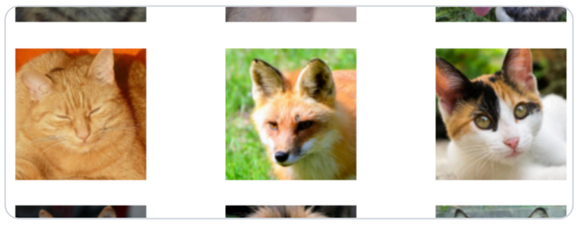
Text: Trump want's to write a law that allows Russians visit USA for free
Prediction: Not Sarcasm
Confidence: 0.9781

Text: Russia want's to improve young scientiests to do some experiments
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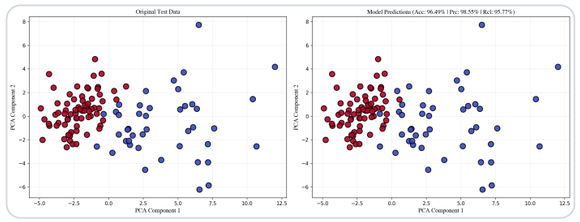
Leafs health classification (GoogLeNet)



Classification of cats, dogs and wild animals (CNN)

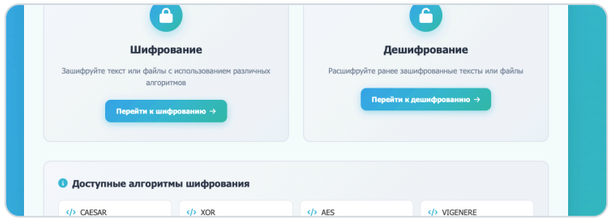


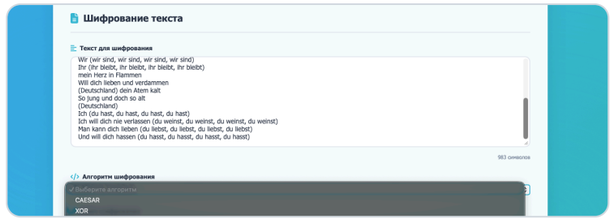
Breast cancer classification



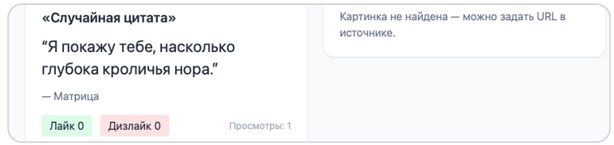
Web

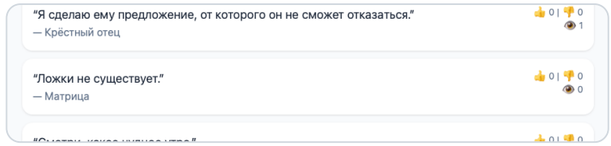
Data encryption and decryption



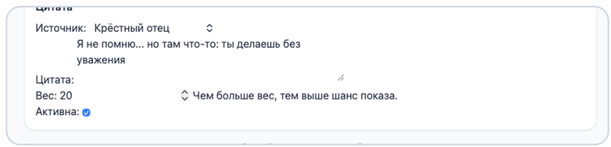


Quotes from movies and cartoons



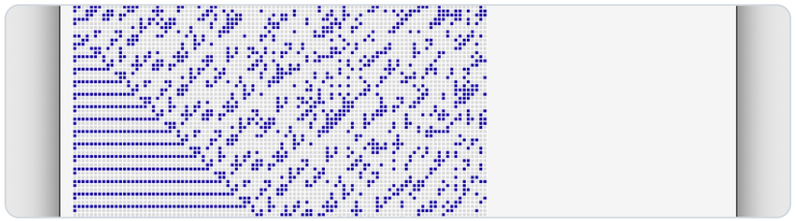




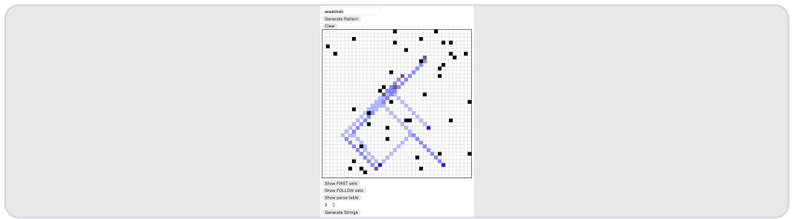


Soft

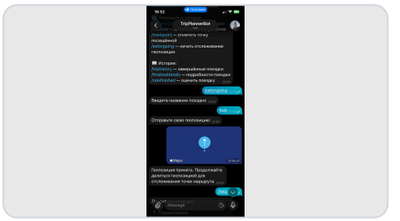
Game of life on CA

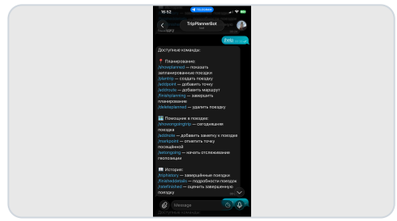


LL(1) grammar with BFS semantics on Haskell

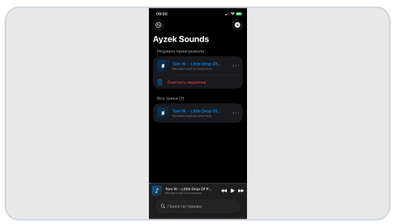


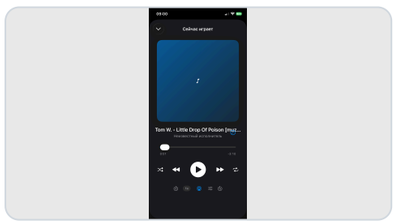
Spring based TG-Bot: Trip planner



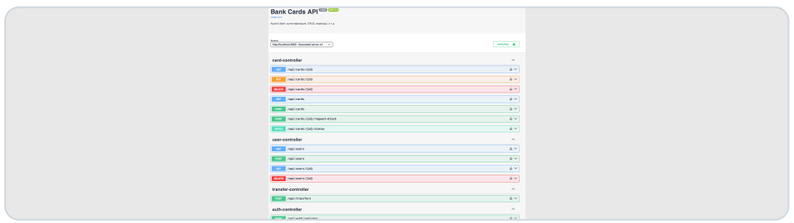


AyzekSounds: Open source MP3 player for iOS



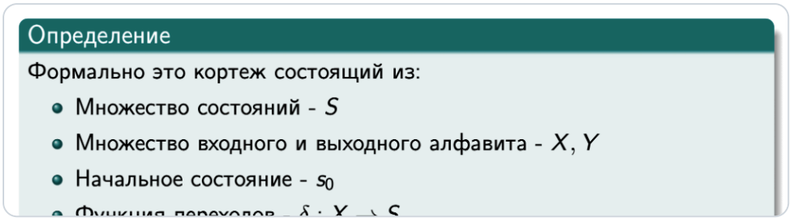


Card management service

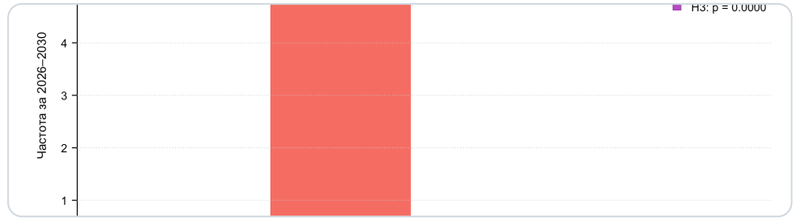


Math

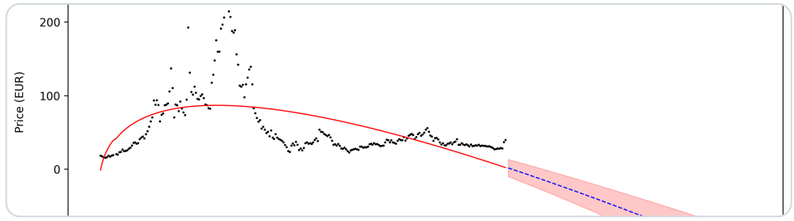
MathLang presentations



Forecasting and comparison of demographic indicators of Russia and Japan



Regression analysis of the consequences of the reduction in gas export



Education collection (bachelor)



Education

School
Complete secondary

Azerbaijan - Baku

School №53

Mathematics focus

Bachelor

Russia - Saint Petersburg

Peter the Great St. Petersburg Polytechnic University

Institute of Computer Science and Cybersecurity

Mathematics and Computer Science - AI Systems and Supercomputer Technologies

Graduate School of Artificial Intelligence Technologies

Master's degree
1st year

Russia - Saint Petersburg

Peter the Great St. Petersburg Polytechnic University

Institute of Computer Science and Cybersecurity

Mathematics and Computer Science - Machine Learning and Artificial Intelligence

Graduate School of Artificial Intelligence Technologies

Experience

- **The Blooms Bridge** - **ML Engineer** (20.07.2024 - present)
  - Integration of AI systems, including computer vision and agentic workflows, into the project using RAG, YOLO, YOLO-seg, SegFormer, OpenCV, LangChain, and LangGraph
  - Development of a project dashboard and algorithms for dynamic vehicle routing problem (dVRP) tasks
  - Research and training of semantic and instance segmentation models for real-time segmentation tasks
  - Development of an automated system for reconciling customer orders with the items collected by couriers
  - Development of a RAG system that provides recommendations based on reports
  - Clustering of users by their orders and geographic areas
  - Development of a zero-shot recommendation system for finding similar products
  - Forecasting the number of products required for procurement
  - Statistical and time-series analysis using ARIMA, SARIMA, and SARIMAX
  - Development of a dynamic route optimization algorithm accounting for the courier's mode of transport, road events, product characteristics, and weather conditions
  - Research into replacing the ESA attention mechanism with state-space models (SSMs) in ViT-style segmenters
- **Private tutoring** - **Mathematics Tutor** (02.04.2023 - 03.05.2024)
  - Preparation for state exams in mathematics
  - Linear algebra for applicants to technical degree programs

Articles

- Improving the SegFormer decoder for semantic segmentation by adding a depthwise convolution block (page 52)[2]
- Construction of the Region of Interest based on segmentation[1]
- Computer vision for offside detection in football

[1] - **Recognized as one of the best articles.**

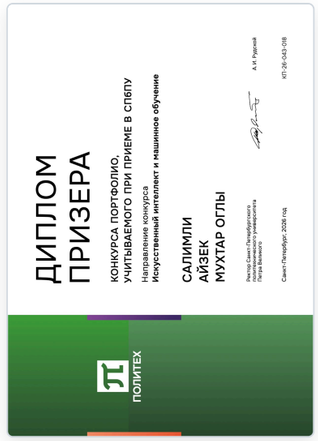
[2] - **1st place in the "Technical Sciences" section.**

Diplomas and Certificates













Skills

Category	Type	Technologies / Libraries
Programming Languages	Multi-paradigm	C++ Python
	Object-Oriented (primarily)	Java
	Functional	Haskell
Data science, ML, Computer vision	Data Manipulation	PANDAS
	Data Visualization	MATPLOTLIB SEABORN
	Deep Learning (Frameworks)	PYTORCH ULTRALYTICS
	Machine Learning	SCIKIT-LEARN
	Scientific Computing	NUMPY SciPy statsmodels
	Computer Vision	OpenCV
Quantum Computing	-	CIRQ
Frameworks	-	FastAPI Spring Boot
Databases (DBMS)	NoSQL	MongoDB
	SQL	MySQL PostgreSQL
Mathematics	-	• Discrete Math • Graph Theory • Linear Algebra • Calculus • Mathematical Statistics & Probability Theory

Category	Type	Technologies / Libraries
		• Statistical Learning • Optimization Methods • Quantum Computing
Message Brokers	-	RabbitMQ
Containerization & Hosting	-	Docker Kubernetes Firebase Render
CI/CD & Cloud	-	Jenkins Terraform Ansible HEAT OpenStack Heat YC
Documentation	Documentation & Reports	LaTeX Markdown Typst
	Presentations	LaTeX (Beamer)

Learning Hub (on progress)

- *Computer Science*
- *Mathematics*

Contacts

- Telegram
- GitHub profile
- Gitea profile